A CRITERION FOR S AND N TO BE QUASI-INVERSE EQUIVALENCES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-S-N (source lines 4231-4245); the proof environment begins on the second line after it and is its own proof, closing at source line 4259.

Editorial note. Neither the statement nor the proof names another labelled result, so the card carries no context span and declares no external pointer; the one sentence delegating the independence check is reproduced as written and recorded in the pairing note.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: a criterion for a pair of exact functors to be quasi-inverse equivalences.

Allowed standard prerequisites (editorial).

- Functors between abelian categories. Exact functors N : A to B and S : B to A between abelian categories, with faithfulness and essential surjectivity.. They are the data of the statement (source lines 4233-4243).
- Quasi-inverse equivalences. Two functors are quasi-inverse when the two composites are isomorphic to the respective identity functors, functorially.. It is the conclusion (source line 4244).
- The natural isomorphism g and exactness. The given isomorphism g : N S to the identity allows elements to be transported, and exactness of N and S makes the transport compatible with the additive structure.. They produce the map of the displayed composite and its independence of the choices (source lines 4248-4257).

Lemma 0.1. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $N : \mathcal{A} \to \mathcal{B}$, and $S : \mathcal{B} \to \mathcal{A}$ be functors. Suppose that*

(1) *the functors S and N are exact,*
(2) *there is an isomorphism $g : N \circ S \to id_{\mathcal{B}}$ to the identity functor of $\mathcal{B}$,*
(3) *N is faithful, and*
(4) *S is essentially surjective.*

Then S and N are quasi-inverse equivalences of categories.

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Proof. It suffices to construct a functorial isomorphism $S(N(A)) \cong A$. To do this choose B and an isomorphism $f : A \to S(B)$. Consider the map

$$f^{-1} \circ g_{S(B)} \circ S(N(f)) : S(N(A)) \to S(N(S(B))) \to S(B) \to A.$$

It is easy to show this does not depend on the choice of f, B and gives the desired isomorphism $S \circ N \to \mathrm{id}_{\mathcal{A}}$. $\square$